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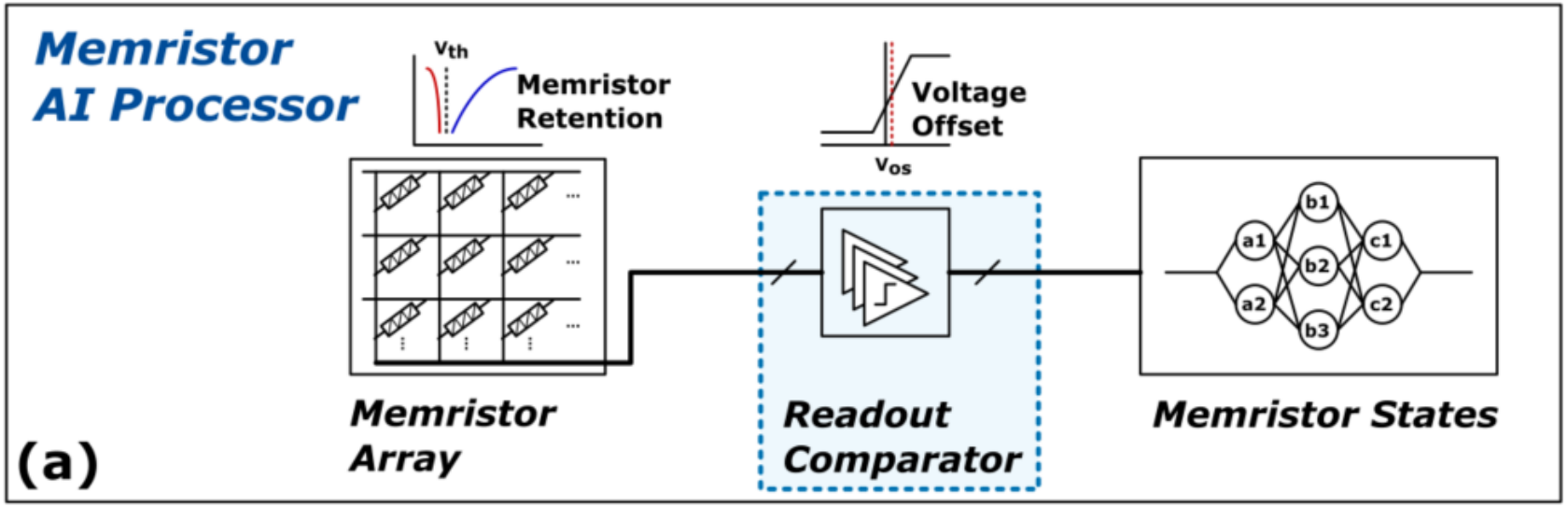
ISCAS 2026

IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS

A Clock-Independent, Time-Domain Rapid Calibration Method for Memristor-based Analog Computing AI Processors

Junzhe Chen, Matthew Schormans, Zhiqiang Que,
Dinesh Pamunuwa, Jiayang Li

Context



- Memristor-based computing
  - Energy efficient
  - Retention
  - Aging

- Function of comparator
  - Accurate and robust readout

Memristor Comparator Challenges

- Small size → large intrinsic input offsets

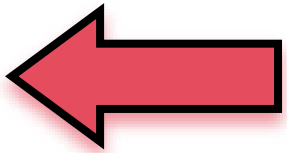
- Large comparator array
  - High consistency (small ΔV_{os})

- Robustness across the whole chip
  - Process
  - Temperature

- Accuracy @ different CLK speed

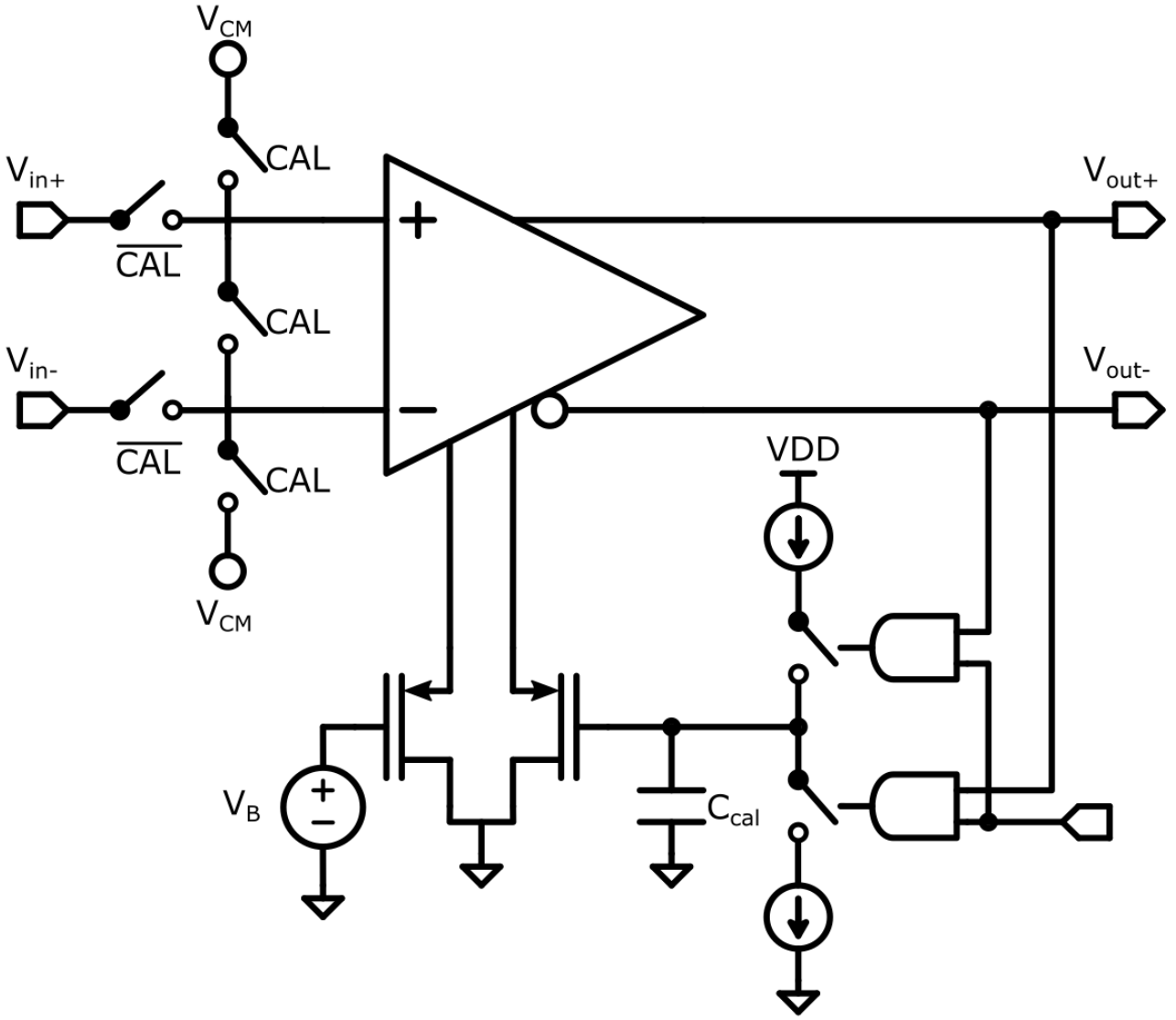
Calibration methods

- Trimming
  - Sensitive to temperature change
- Storage
  - Recalibration
- Dynamic
  - Calibration speed
  - Calibration accuracy



Prior Arts: Miyahara [1]

- Output → charge pump → input
- counter offset voltage @ input
- Accuracy depends on
  - CLK speed
  - C_{cal} value



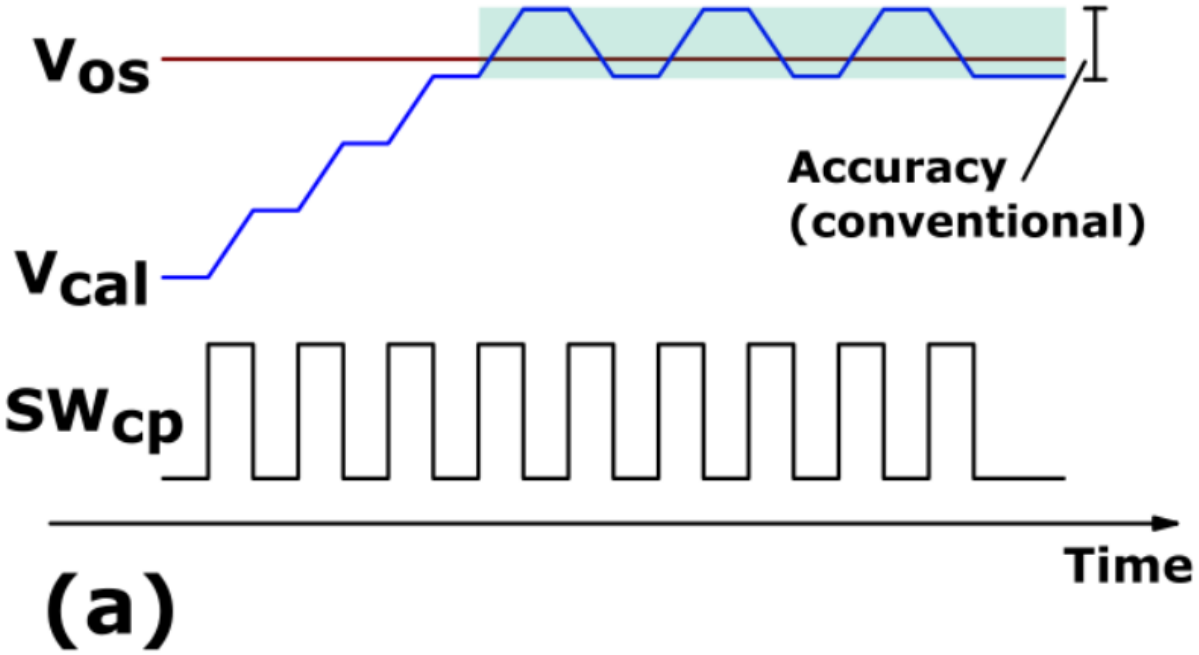
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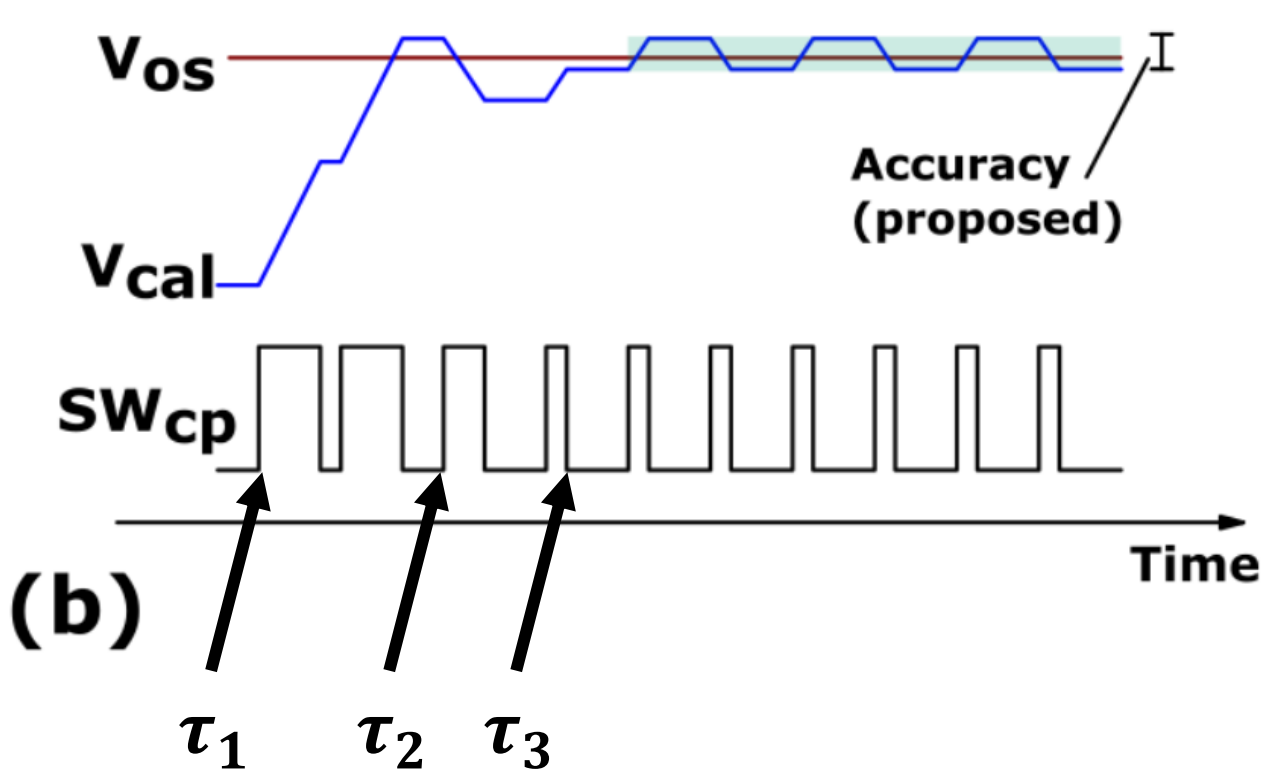
Problem: Miyahara

- Calibration time – Accuracy tradeoff
- Slower CLK → worse accuracy

Solution

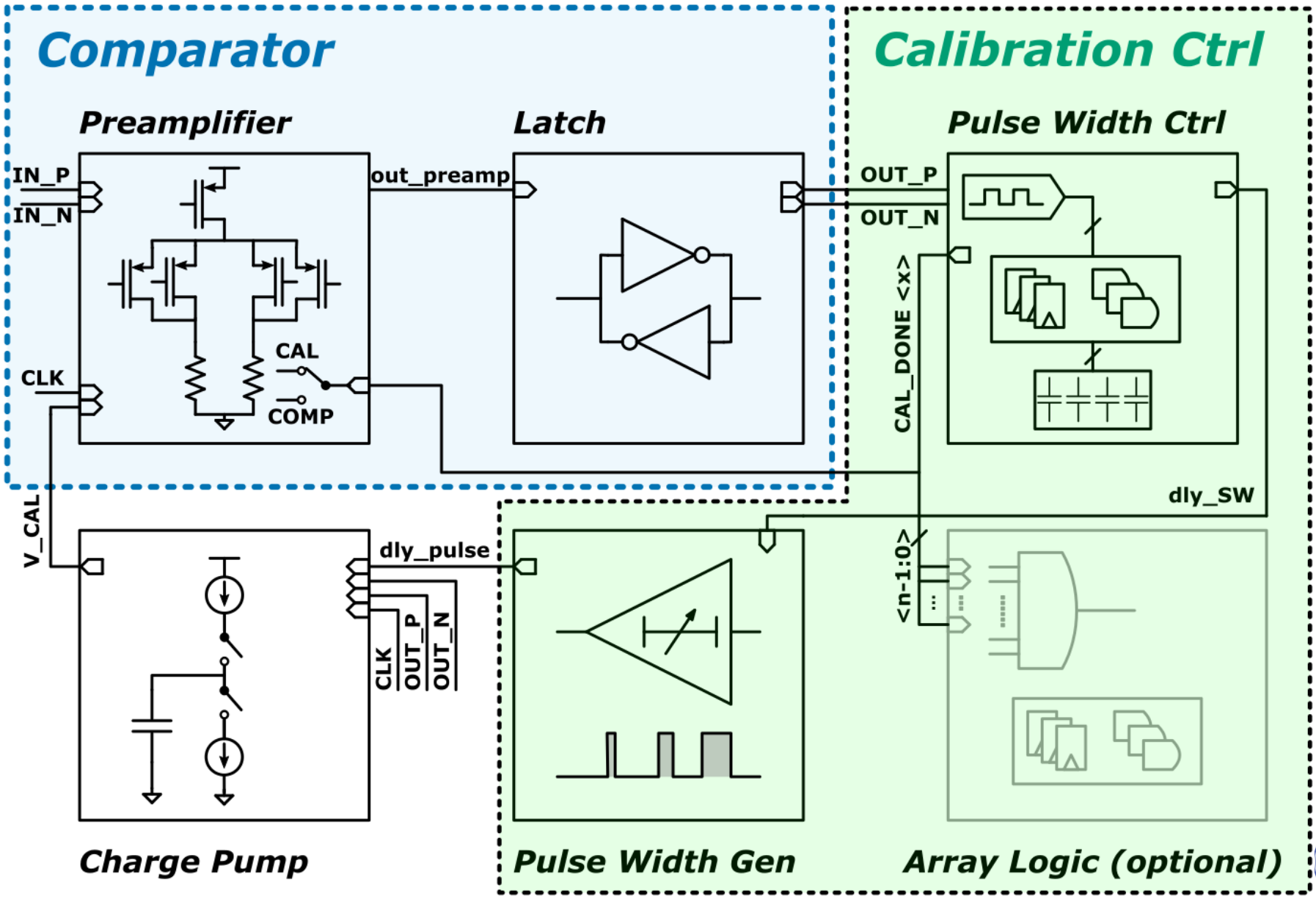
Output-controlled CP → Set width pulse





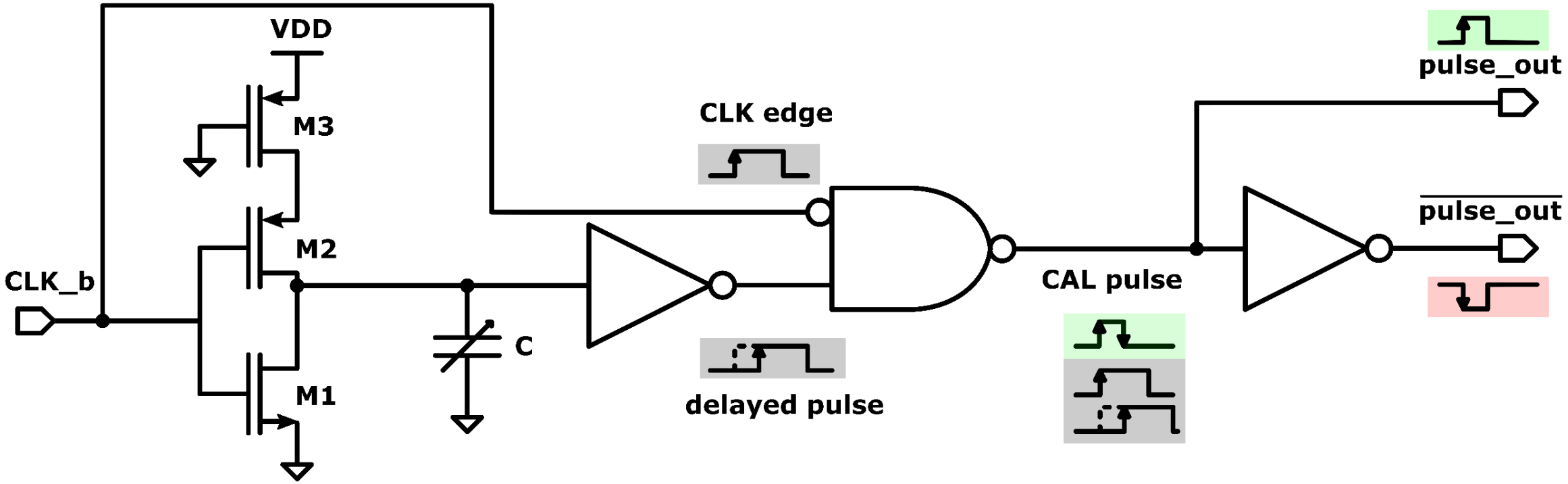
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Circuit block diagram



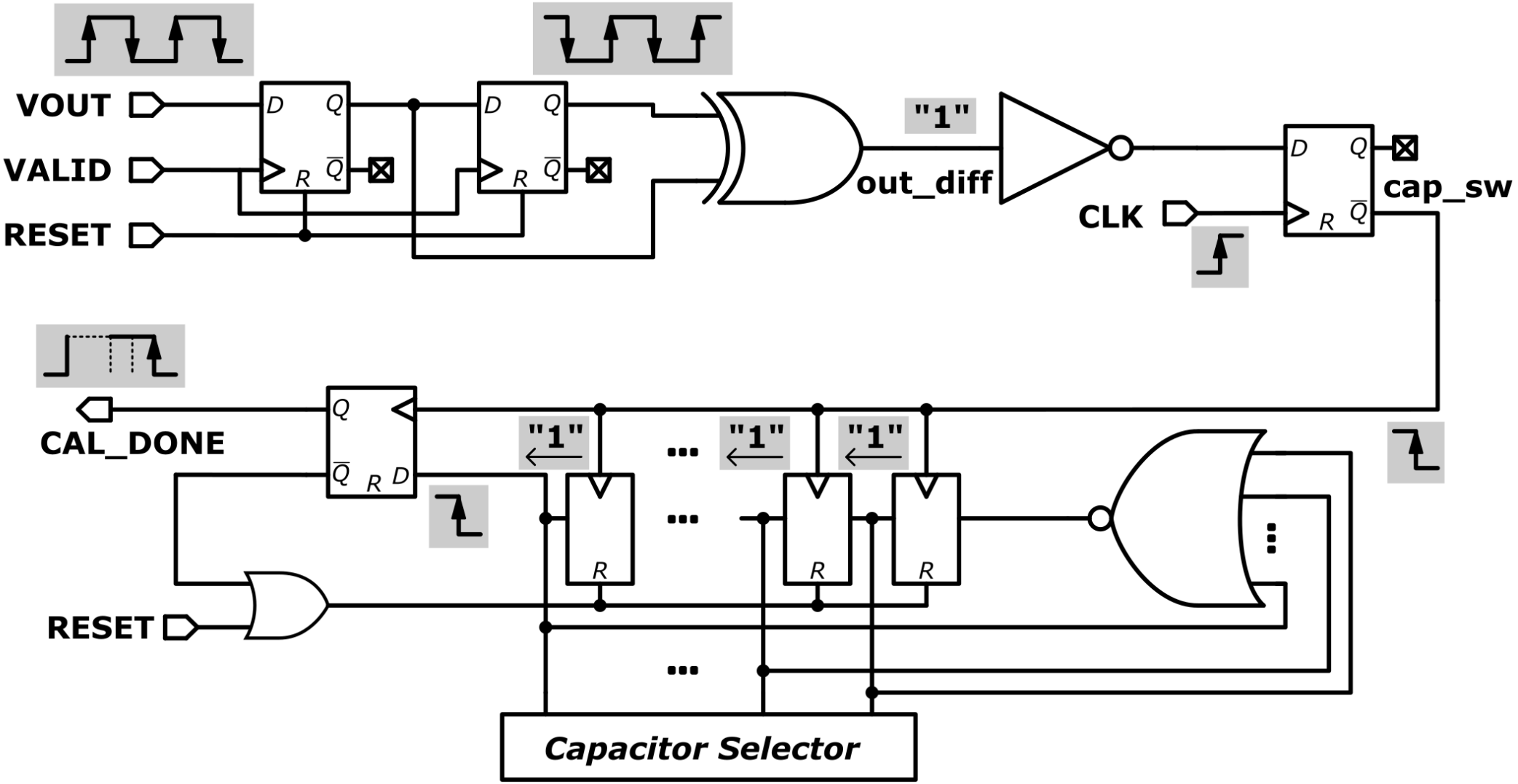
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Pulse Generator



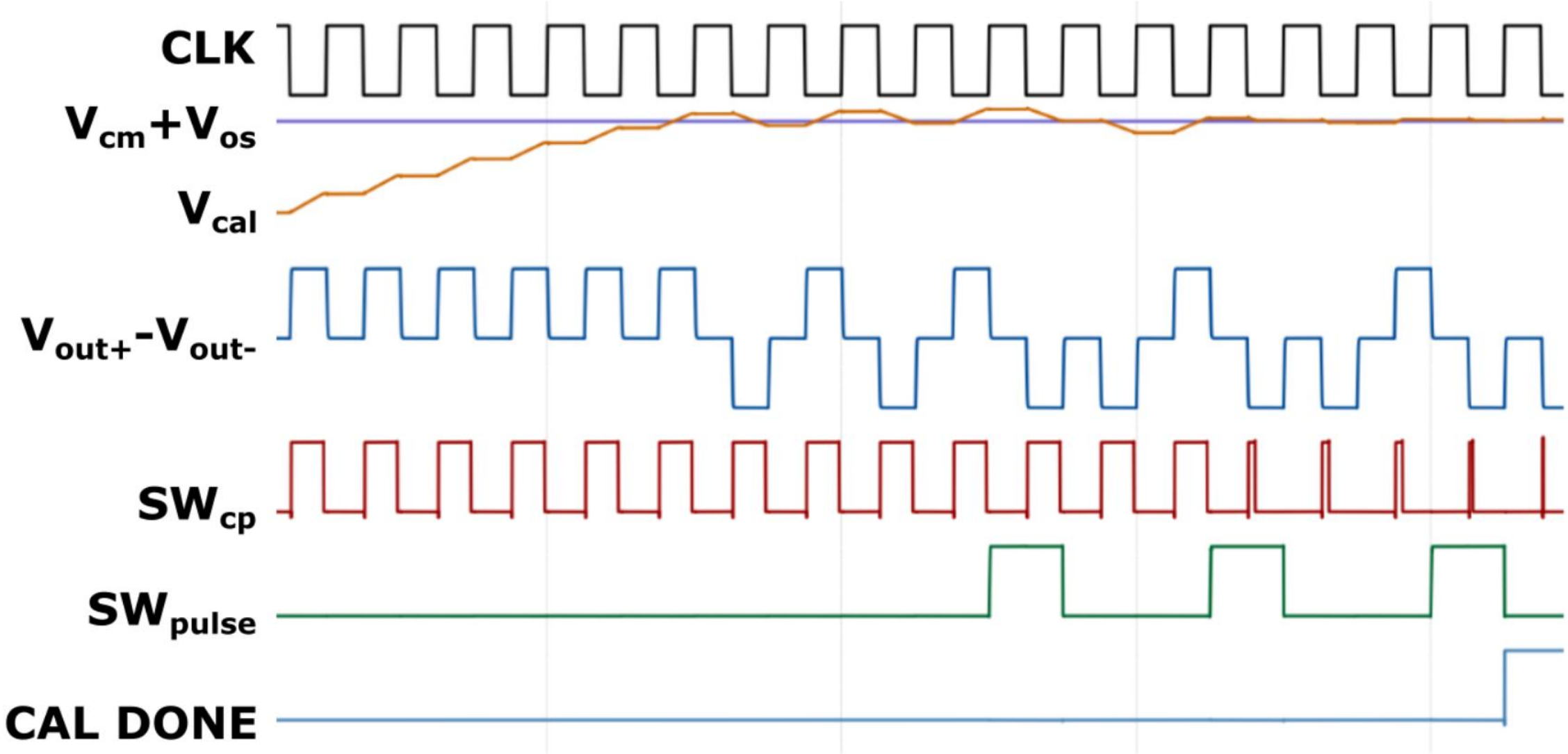
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Control Block



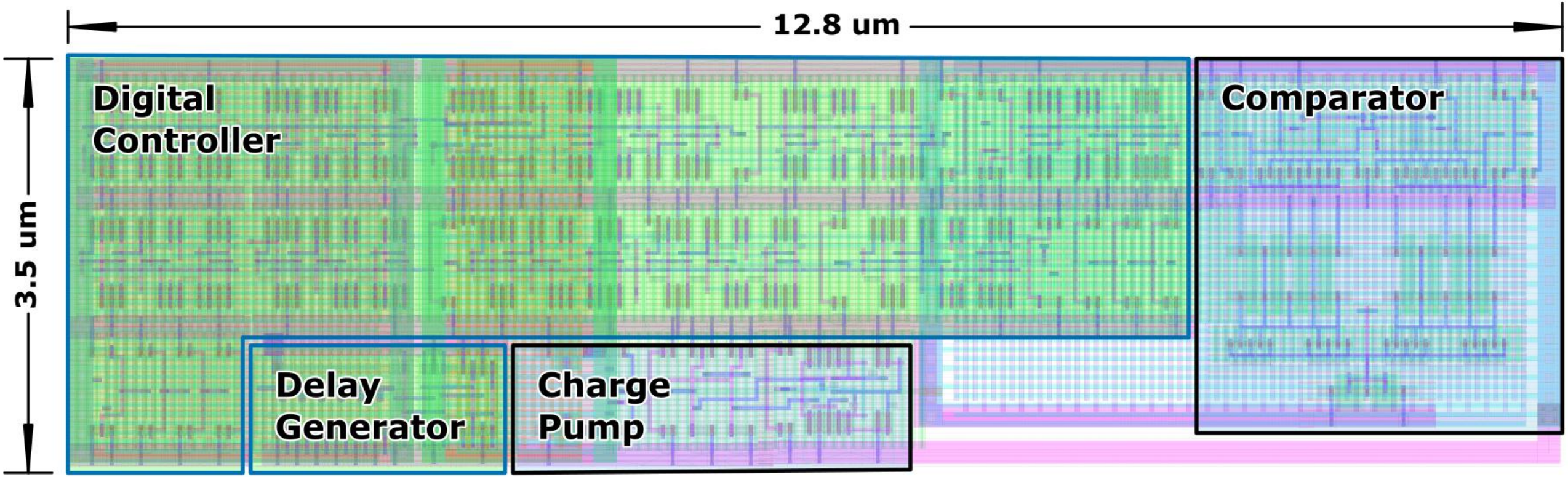
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Timing Diagram

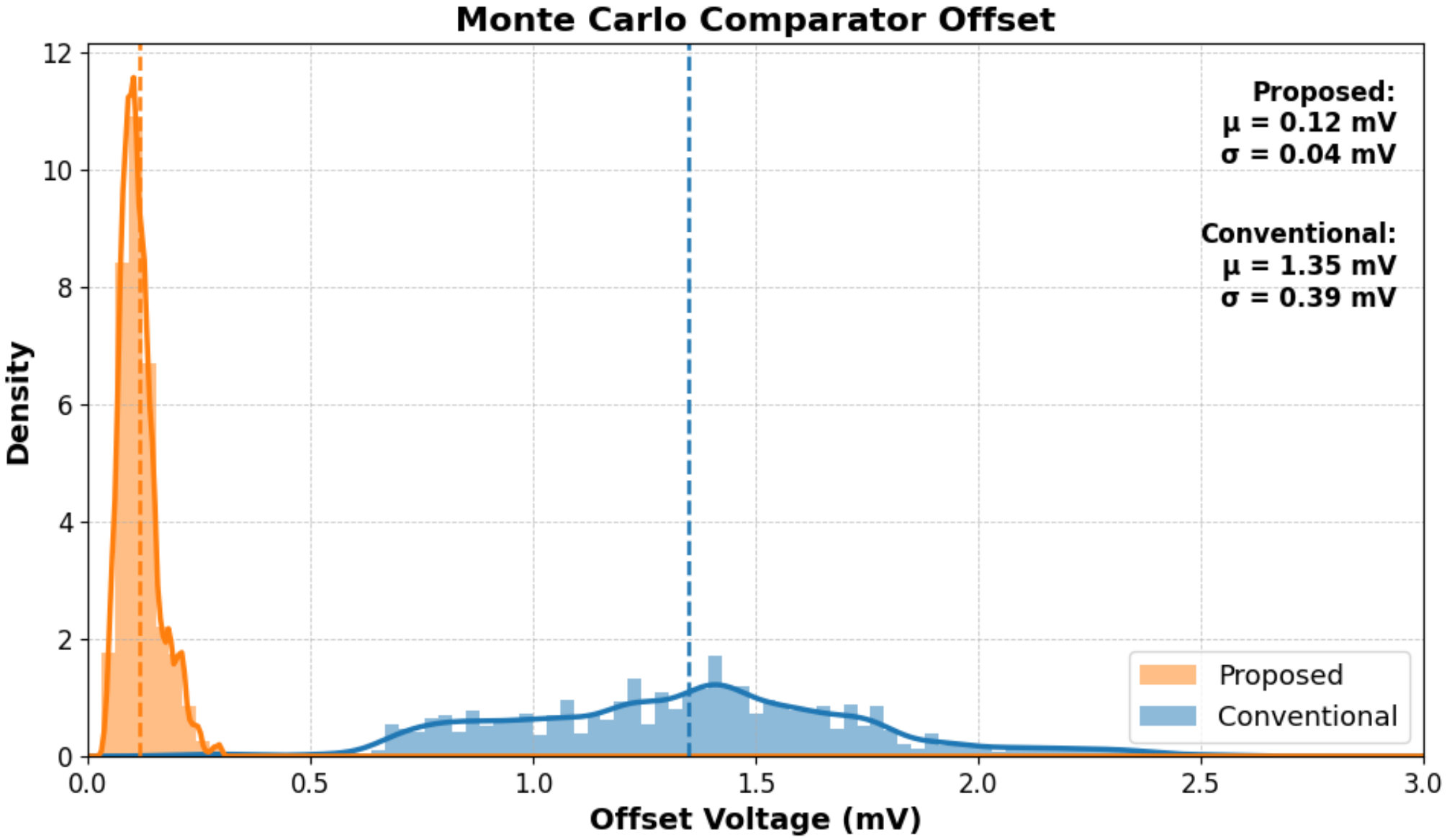


Results: layout

- **Process:** GF 12 nm
- **Area (single unit):** 44.8 μm^2

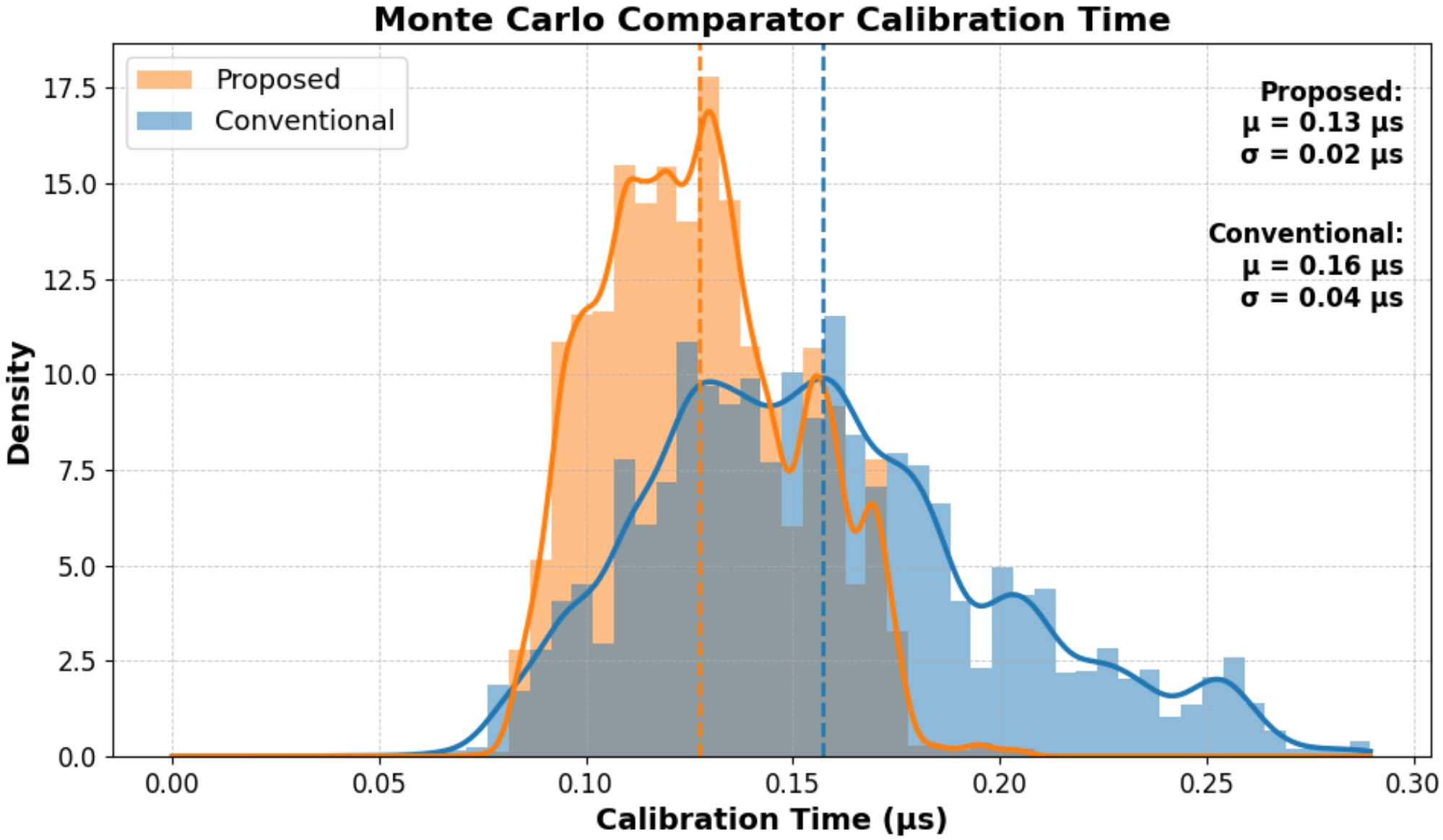


Results: Offsets



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Results: Calibration Time



Results: Comparison

- Low offset error compared to other dynamic calibration schemes

- Low power, thanks to advanced nodes

	This work	ISCAS'25 [2]	ISCAS'25 [3]	ISCAS'20 [4]	ASSCC'11 [1]
Process	**12 nm**	12 nm	28 nm	14 nm	90 nm
Operation frequency	**50–500 MHz**	1.5 GHz	4 GHz	1 GHz	250 MHz
Offset	**120 µV**	1.9 mV	136 µV	259 µV	1.3 mV
Power	**10 µW**	240 µW	220 µW	213 µW	40 µW

Thanks for your attention

References

1. Miyahara et al. "A low-noise self-calibrating dynamic comparator for high-speed ADCs," *2008 IEEE Asian Solid-State Circuits Conference*
2. Yang et al.“A high-speed high-precision comparator with offset calibration for sha less pipeline adcs,” in 2025 IEEE International Symposium on Circuits and Systems (ISCAS)
3. Xu et al. “A low-noise high-precision dual-mode offset calibration for high-speed dynamic comparators,” in 2025 IEEE International Symposium on Circuits and Systems (ISCAS)
4. Lee et al. “An effective transconductance controlled offset calibration for dynamic comparators,” in Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS)